

RoCE诊断工具

连通测试	
查看roce网卡	
连通测试	
PFC ECN测试	
带宽压测	
观察cnp拥塞和pause帧	

连通测试

完成roce对接包之后，在节点上可以查看支持roce的ib设备，使用ib_send_bw完成连通测试。

查看roce网卡

```
1 $ ibdev2netdev
2 mlx5_0 port 1 ==> enp61s0f3 (Up)
3 mlx5_1 port 1 ==> enp61s0f2 (Up)
```

连通测试

node-2启动服务端

```
1 ib_send_bw -R -d mlx5_2
```

node-1启动客户端

```
1 ib_send_bw -R -d mlx5_2 9.9.8.10
```

PFC ECN测试

PFC和ECN分别完成入口和出口的拥塞控制，PFC和ECN都需要交换机参与，在网卡侧可以观察到暂停帧和CNP拥塞通知报文。

带宽压测

- node-2

```
1 for i in {0..7} ; do taskset -c $i ib_send_bw -R -d mlx5_2 -F --report_gbits -f 2 -D 300 -p $((10000+i)) & done
```

- node-1

```
1 for i in {0..7} ; do taskset -c $i ib_send_bw -R -d mlx5_2 -F --report_gbits -f 2 -D 300 9.9.8.10 -p $((10000+i)) & done | grep 65536 | awk '{sum+=$4} END {print sum}'
```

观察cnp拥塞和pause帧

```
1 cat > hw_counter.sh << 'EOF'
2 #!/bin/bash
3
4 port=$1
5
6 if [ "$port"x == ""x ]; then
7     echo "Usage: $0 <port>"
8     echo ""
9     ibdev2netdev
10    exit
```

```
11 fi
12
13 ibpath=$(find /sys/class/net/$port/device/infiniband -maxdepth 1 -mindepth 1)
14 hw_counter="$ibpath/ports/1/hw_counters"
15
16 echo "# HW Counter:"
17 for c in np_cnp_sent rp_cnp_handled; do
18     echo $c $(cat $hw_counter/$c)
19 done
20
21 echo "# Statis:"
22 ethtool -S $port | grep -e prio3 -e prio6
23
24 EOF
25
26 chmod +x hw_counter.sh
```

```
1 $ ./hw_counter.sh enp2s0f0
```

```
1 $ watch './hw_counter.sh enp2s0f0'
2 # HW Counter:
3 np_cnp_sent 6316885
4 rp_cnp_handled 0
5 # Statis:
6 rx_prio3_bytes: 5550382400650
7 rx_prio3_packets: 1345419587
8 rx_prio3_discards: 0
9 tx_prio3_bytes: 11481188652
10 tx_prio3_packets: 29199658
11 rx_prio6_bytes: 76892092
12 rx_prio6_packets: 1165026
13 rx_prio6_discards: 7
14 tx_prio6_bytes: 517984570
15 tx_prio6_packets: 6316885
16 rx_prio3_pause: 0
17 rx_prio3_pause_duration: 0
18 tx_prio3_pause: 0
19 tx_prio3_pause_duration: 0
20 rx_prio3_pause_transition: 0
21 rx_prio3_buf_discard: 0
22 rx_prio3_cong_discard: 0
23 rx_prio3_marked: 0
24 rx_prio6_buf_discard: 0
25 rx_prio6_cong_discard: 0
26 rx_prio6_marked: 0
```